

YEAR 7 MATHEMATICS

Chapter 3

Whole numbers

WORKSHEETS

3

How to use this booklet

1 There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

2 Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

WE DO **Top of each column.** Your teacher works the examples on the board. **Copy every step** into the space.

YOU DO **The rest of the column.** Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

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In this chapter you will

- round and estimate
- multiply and divide whole numbers
- use divisibility tests
- use long division and Roman numerals
- use index notation, square roots and cube roots
- find prime numbers and prime factors
- find the HCF and LCM of two numbers

About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the tentacled edge of a small copy of the whole set, out on the antenna near -1.77 . Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

Easy

Questions 1–19

Round to the place given.
Look at the digit to its right:
5 or more, round up.



WE DO With your teacher: copy the working.

E1 7 319 to the nearest 100 _____

E2 89 310 to the nearest 1 000 _____

YOU DO Your turn.

1 4 217 to the nearest 100 4 200

2 313 to the nearest 10 _____

3 7 047 to the nearest 100 _____

4 4 481 to the nearest 100 _____

5 275 to the nearest 10 _____

6 2 414 to the nearest 100 _____

7 29 674 to the nearest 1 000 _____

8 641 to the nearest 10 _____

9 4 488 to the nearest 100 _____

10 915 to the nearest 10 _____

11 4 361 to the nearest 100 _____

12 4 857 to the nearest 100 _____

13 692 to the nearest 10 _____

14 6 748 to the nearest 100 _____

15 951 to the nearest 10 _____

16 4 510 to the nearest 100 _____

17 11 to the nearest 10 _____

18 88 923 to the nearest 1 000 _____

19 177 to the nearest 10 _____

Medium

Questions 20–41



Estimate: round each number to its first digit, then work it out.

WE DO With your teacher: copy the working.

E1 450×12 **E2** $86 + 27$

YOU DO Your turn.

20 750×43 **21** $26 + 45$
 $\approx 800 \times 40$ $\approx 30 + 50$

$\approx 32\ 000$ $\approx \underline{\hspace{2cm}}$

22 480×28 **23** $980 + 95$
 $\approx \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$ _____

$\approx \underline{\hspace{2cm}}$ _____

24 82×74 **25** $700 + 26$

26 $26 + 80$ **27** 17×58

28 $610 + 18$ **29** $150 + 16$

30 900×67 **31** 800×32

32 $63 + 96$ **33** $200 + 17$

34 130×59 **35** 430×45

36 $910 + 79$ **37** 540×29

38 460×38 **39** $73 + 44$

40 970×84 **41** $620 + 71$

Challenging

Questions 42–49

Estimate, then answer the question. Show your rounding.



WE DO With your teacher: copy the working.

E1 29 students each pay \$16 for an excursion. Estimate the total.

E2 495 people share 36 buses. About how many are on each bus?

YOU DO Your turn.

42 38 students each pay \$18 for an excursion. Estimate the total.

$\approx 40 \times 20$

$\approx \underline{\hspace{2cm}}$

43 703 people share 33 buses. About how many are on each bus?

$\approx \underline{\hspace{2cm}} \div \underline{\hspace{2cm}}$

$\approx \underline{\hspace{2cm}}$

44 44 students each pay \$27 for an excursion. Estimate the total.

45 782 people share 29 buses. About how many are on each bus?

46 21 students each pay \$24 for an excursion. Estimate the total.

47 817 people share 38 buses. About how many are on each bus?

48 38 students each pay \$19 for an excursion. Estimate the total.

49 613 people share 24 buses. About how many are on each bus?

Easy

Questions 1–25

Know your times tables.

WE DO With your teacher: copy the working.

E1 10×10

E2 12×9

YOU DO Your turn.

1 10×5

50

2 6×12

3 5×11

4 6×3

5 9×6

6 3×10

7 5×9

8 3×4

9 6×9

10 12×7

11 11×4

12 10×8

Multiply by 10, 100 or 1000: move each digit left one place for each zero.

WE DO With your teacher: copy the working.

E3 67×100

YOU DO Your turn.

13 $55 \times 1\,000$

55 000

14 52×10

15 55×10

16 24×10

17 31×100

18 $75 \times 1\,000$

19 $71 \times 1\,000$

20 88×10

21 56×10

22 31×10

23 95×100

24 $32 \times 1\,000$

25 12×100

Medium

Questions 26–41

Split the number to multiply in your head: $34 \times 6 = 30 \times 6 + 4 \times 6$.

WE DO With your teacher: copy the working.

E1 54×4

E2 33×5

YOU DO Your turn.

26 33×4

$= 30 \times 4 + 3 \times 4$

$= 120 + 12$

$= 132$

28 39×9

$= ___ \times ___ + ___ \times ___$

$= ___ + ___$

$= ___$

30 74×3

$= ___ \times ___ + ___ \times ___$

$= ___ + ___$

$= ___$

32 38×6

$= ___ \times ___ + ___ \times ___$

$= ___ + ___$

$= ___$

34 79×8

$= ___ \times ___ + ___ \times ___$

$= ___ + ___$

$= ___$

36 17×3

$= ___ \times ___ + ___ \times ___$

$= ___ + ___$

$= ___$

38 48×3

$= ___ \times ___ + ___ \times ___$

$= ___ + ___$

$= ___$

40 64×3

$= ___ \times ___ + ___ \times ___$

$= ___ + ___$

$= ___$

27 88×8

$= 80 \times 8 + 8 \times 8$

$= 640 + 64$

$= ___$

29 75×6

$= ___ \times ___ + ___ \times ___$

$= ___ + ___$

$= ___$

31 49×3

$= ___ \times ___ + ___ \times ___$

$= ___ + ___$

$= ___$

33 46×4

$= ___ \times ___ + ___ \times ___$

$= ___ + ___$

$= ___$

35 74×4

$= ___ \times ___ + ___ \times ___$

$= ___ + ___$

$= ___$

37 13×6

$= ___ \times ___ + ___ \times ___$

$= ___ + ___$

$= ___$

39 76×5

$= ___ \times ___ + ___ \times ___$

$= ___ + ___$

$= ___$

41 52×4

$= ___ \times ___ + ___ \times ___$

$= ___ + ___$

$= ___$

Challenging

Questions 42–48

Multiply a 3-digit number by a 2-digit number. Split the second number.

WE DO With your teacher: copy the working.

E1 300×15

E2 326×23

YOU DO Your turn.

42 308×24

$308 \times 20 = 6\,160$

$308 \times 4 = 1\,232$

$___ + ___ = ___$

43 466×34

$___ \times ___ = ___$

$___ \times ___ = ___$

$___ + ___ = ___$

44 249×25

$___ \times ___ = ___$

$___ \times ___ = ___$

$___ + ___ = ___$

45 286×33

$___ \times ___ = ___$

$___ \times ___ = ___$

$___ + ___ = ___$

46 314×17

$___ \times ___ = ___$

$___ \times ___ = ___$

$___ + ___ = ___$

47 127×26

$___ \times ___ = ___$

$___ \times ___ = ___$

$___ + ___ = ___$

48 122×17

$___ \times ___ = ___$

$___ \times ___ = ___$

$___ + ___ = ___$

Easy

Questions 1–26

Know your division facts.



WE DO

With your teacher: copy the working.

E1 $77 \div 7$ _____

E2 $36 \div 4$ _____

YOU DO Your turn.

1 $40 \div 8$ _____

5

2 $81 \div 9$ _____

3 $45 \div 5$ _____

4 $35 \div 5$ _____

5 $110 \div 11$ _____

6 $132 \div 12$ _____

7 $33 \div 11$ _____

8 $66 \div 6$ _____

9 $54 \div 9$ _____

10 $108 \div 12$ _____

11 $48 \div 4$ _____

12 $32 \div 4$ _____

13 $66 \div 11$ _____

Divide by 10, 100 or 1000.



WE DO

With your teacher: copy the working.

E3 $120 \div 10$ _____

YOU DO Your turn.

14 $23\,000 \div 1\,000$ _____

23

15 $8\,800 \div 100$ _____

16 $50\,000 \div 1\,000$ _____

17 $780 \div 10$ _____

18 $100 \div 10$ _____

19 $490 \div 10$ _____

20 $71\,000 \div 1\,000$ _____

21 $83\,000 \div 1\,000$ _____

22 $9\,300 \div 100$ _____

23 $17\,000 \div 1\,000$ _____

24 $950 \div 10$ _____

25 $22\,000 \div 1\,000$ _____

26 $90 \div 10$ _____

Medium

Questions 27–42



Split the number into parts you can divide easily.

WE DO

With your teacher: copy the working.

E1 $108 \div 6$ _____

E2 $342 \div 6$ _____

YOU DO Your turn.

27 $344 \div 4$ _____

$= 320 \div 4 + 24 \div 4$

$= 80 + 6$

$= 86$

28 $198 \div 3$ _____

$= 180 \div 3 + 18 \div 3$

$= 60 + 6$

$= \underline{\hspace{1cm}}$

29 $344 \div 8$ _____

$= \underline{\hspace{1cm}} \div \underline{\hspace{1cm}} + \underline{\hspace{1cm}}$

$= \underline{\hspace{1cm}} + \underline{\hspace{1cm}}$

$= \underline{\hspace{1cm}}$

30 $528 \div 8$ _____

31 $297 \div 3$ _____

32 $656 \div 8$ _____

33 $498 \div 6$ _____

34 $333 \div 9$ _____

35 $819 \div 9$ _____

36 $63 \div 3$ _____

37 $39 \div 3$ _____

38 $310 \div 5$ _____

39 $396 \div 6$ _____

40 $348 \div 4$ _____

41 $138 \div 6$ _____

42 $306 \div 6$ _____

Challenging

Questions 43–48

Use short division. Write the remainder, then answer the question.



WE DO

With your teacher: copy the working.

E1 276 lollies are shared by 5 people. How many each, and how many are left over?

E2 136 lollies are shared by 9 people. How many each, and how many are left over?

YOU DO Your turn.

43 157 lollies are shared by 9 people. How many each, and how many are left over?

$157 \div 9 = 17 \text{ r } 4$

___ each, ___ left over

44 413 lollies are shared by 5 people. How many each, and how many are left over?

$\underline{\hspace{1cm}} \div \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ r } \underline{\hspace{1cm}}$

___ each, ___ left over

45 369 lollies are shared by 8 people. How many each, and how many are left over?

46 211 lollies are shared by 6 people. How many each, and how many are left over?

47 463 lollies are shared by 7 people. How many each, and how many are left over?

48 418 lollies are shared by 7 people. How many each, and how many are left over?

Easy

Questions 1–25

Is the number divisible by the number in brackets? Write yes or no.

WE DO With your teacher: copy the working.

E1 2 298 (10)

E2 6 988 (2)

YOU DO Your turn.

1 8 958 (5)

no

2 4 241 (5)

3 8 802 (5)

4 4 576 (5)

5 3 368 (2)

6 945 (10)

7 1 644 (10)

8 4 414 (5)

9 661 (10)

10 458 (10)

11 4 370 (10)

12 7 861 (2)

Find the digit sum. Is the number divisible by 3?

WE DO With your teacher: copy the working.

E3 7 427

YOU DO Your turn.

13 3 246

15, yes

14 1 121

15 910

16 7 697

17 8 126

18 6 711

19 2 326

20 8 527

21 3 191

22 780

23 2 186

24 3 559

25 2 872

Medium

Questions 26–34

Use the test. Show the check, then write yes or no.

WE DO With your teacher: copy the working.

E1 Is 5 211 divisible by 9?

E2 Is 7 260 divisible by 4?

E3 Is 5 978 divisible by 6?

YOU DO Your turn.

26 Is 3 088 divisible by 9?

digit sum: $3 + 0 + 8 + 8 = 19$

19 is not divisible by 9, so no

27 Is 6 989 divisible by 4?

last two digits: 89

89 is not divisible by 4, so no

28 Is 8 566 divisible by 6?

even? yes; digit sum 29

no: not divisible by both 2 and 3

29 Is 2 255 divisible by 9?

30 Is 4 923 divisible by 4?

31 Is 6 096 divisible by 6?

32 Is 9 656 divisible by 9?

33 Is 5 734 divisible by 4?

34 Is 8 838 divisible by 6?

Challenging

Questions 35–39

Find the missing digit so that the number is divisible as shown.

WE DO With your teacher: copy the working.

E1 127□ is divisible by 3. Find every digit that fits the box.

E2 651□ is divisible by 9. Find every digit that fits the box.

YOU DO Your turn.

35 986□ is divisible by 3. Find every digit that fits the box.

$9 + 8 + 6 = 23$

$23 + \square$ must be a multiple of 3

$\square = _, _, _$

36 922□ is divisible by 3. Find every digit that fits the box.

$9 + 2 + 2 + \square = _$

$_ + \square$ must be a multiple of 3

$\square = _, _, _$

37 747□ is divisible by 9. Find every digit that fits the box.

38 295□ is divisible by 9. Find every digit that fits the box.

39 489□ is divisible by 9. Find every digit that fits the box.

Easy

Questions 1–25

Know your multiples. You need them for long division.

WE DO With your teacher: copy the working.

E1 22×7 _____

E2 20×9 _____

YOU DO Your turn.

1 24×4 _____ **96**

2 22×4 _____

3 11×6 _____

4 23×5 _____

5 25×8 _____

6 19×2 _____

7 13×4 _____

8 24×3 _____

9 17×8 _____

10 16×3 _____

11 12×7 _____

12 21×6 _____

How many times does the first number go into the second? Write the remainder.

WE DO With your teacher: copy the working.

E3 15 into 35 _____

YOU DO Your turn.

13 17 into 104 _____ **6 r 2**

14 13 into 126 _____

15 25 into 103 _____

16 14 into 34 _____

17 24 into 211 _____

18 13 into 124 _____

19 16 into 116 _____

20 20 into 180 _____

21 25 into 89 _____

22 20 into 155 _____

23 17 into 75 _____

24 14 into 59 _____

25 16 into 76 _____

Medium

Questions 26–32

Use long division. Divide, multiply, subtract, bring down.

WE DO With your teacher: copy the working.

E1 $504 \div 21$ _____

E2 $1\,050 \div 25$ _____

YOU DO Your turn.

26 $1\,608 \div 24$ _____

160 $\div 24 = 6 \text{ r } 16$

bring down: $168 \div 24 = 7$

answer: **67**

27 $360 \div 15$ _____

36 $\div 15 = 2 \text{ r } 6$

bring down: $60 \div 15 = 4$

answer: _____

28 $912 \div 16$ _____

____ $\div$ ____ = ____ **r** ____

bring down: ____ $\div$ ____ = ____

answer: _____

29 $888 \div 24$ _____

30 $676 \div 26$ _____

31 $429 \div 13$ _____

32 $720 \div 12$ _____

Challenging

Questions 33–36

Use long division, then answer the question.

WE DO With your teacher: copy the working.

E1 1211 students are put into teams of 33. How many full teams, and how many students are left over?

E2 1015 students are put into teams of 36. How many full teams, and how many students are left over?

YOU DO Your turn.

33 609 students are put into teams of 41. How many full teams, and how many students are left over?

609 $\div 41$ _____

= 14 r 35

____ full teams, ____ left over

34 703 students are put into teams of 26. How many full teams, and how many students are left over?

____ $\div$ ____

= ____ r ____

____ full teams, ____ left over

35 522 students are put into teams of 26. How many full teams, and how many students are left over?

36 1492 students are put into teams of 44. How many full teams, and how many students are left over?

Easy

Questions 1–25

Write each Roman numeral as a number. I = 1, V = 5, X = 10, L = 50, C = 100.

WE DO With your teacher: copy the working.

E1 XXIII

E2 XL

YOU DO Your turn.

1 XXXV

35

2 XXX

3 LXII

4 IX

5 LIV

6 LXXXVII

7 LVI

8 LXX

9 XCVIII

10 LXIV

11 LXXV

12 XXXVI

Write each number in Roman numerals.

WE DO With your teacher: copy the working.

E3 27

YOU DO Your turn.

13 42

XLII

14 24

15 67

16 2

17 35

18 32

19 81

20 41

21 53

22 14

23 77

24 50

25 20

Medium

Questions 26–35

Split the number into place values, then write each part in Roman numerals.

WE DO With your teacher: copy the working.

E1 490

E2 652

YOU DO Your turn.

26 843

800 + 40 + 3

DCCC + XL + III = DCCCXLIII

27 694

600 + 90 + 4

28 780

29 831

30 319

31 296

32 258

33 455

34 409

35 478

Challenging

Questions 36–42

Write each year in Roman numerals. Show the parts.

WE DO With your teacher: copy the working.

E1 the year 1849

E2 the year 2003

YOU DO Your turn.

36 the year 1906

1000 + 900 + 6

M + CM + VI

37 the year 1993

38 the year 1861

39 the year 1949

40 the year 1956

41 the year 1915

42 the year 1936

Easy

Questions 1–25

Write in index notation:
count how many times the
number is multiplied.

WE DO With your teacher: copy the
working.

E1 5×5 _____

E2 9×9 _____

YOU DO Your turn.

1 4×4 4^2

2 $2 \times 2 \times 2 \times 2 \times 2$ _____

3 $4 \times 4 \times 4$ _____

4 7×7 _____

5 $4 \times 4 \times 4 \times 4 \times 4 \times 4$ _____

6 $7 \times 7 \times 7$ _____

7 $2 \times 2 \times 2 \times 2$ _____

8 $7 \times 7 \times 7 \times 7 \times 7 \times 7$ _____

9 $3 \times 3 \times 3$ _____

10 $3 \times 3 \times 3 \times 3$ _____

11 $6 \times 6 \times 6$ _____

12 $2 \times 2 \times 2$ _____

Evaluate each power.

WE DO With your teacher: copy the
working.

E3 8^2 _____

YOU DO Your turn.

13 7^2 49

14 2^5 _____

15 2^3 _____

16 5^2 _____

17 2^2 _____

18 9^2 _____

19 4^3 _____

20 10^2 _____

21 5^3 _____

22 3^4 _____

23 4^2 _____

24 3^5 _____

25 6^2 _____

Medium

Questions 26–32

Write the power as a
product, then evaluate it.

WE DO With your teacher: copy the
working.

E1 5^3 _____

E2 4^3 _____

YOU DO Your turn.

26 2^3
 $= 2 \times 2 \times 2$
 $= 8$

27 6^3
 $= 6 \times 6 \times 6$
 $= ______$

28 3^4
 $= ______ \times ______ \times ______ \times ______$
 $= ______$

29 2^4 _____

30 3^3 _____

31 3^5 _____

32 2^5 _____

Challenging

Questions 33–42

Evaluate each power first,
then do the operation.

WE DO With your teacher: copy the
working.

E1 $5^2 \times 5^2$ _____

E2 $4^2 \times 2^2$ _____

YOU DO Your turn.

33 $3^2 + 5^2$
 $= 9 + 25$
 $= ______$

34 $4^2 + 2^2$
 $= ______ + ______$
 $= ______$

35 $5^3 \times 4^2$ _____

36 $2^2 + 3^2$ _____

37 $4^3 + 5^2$ _____

38 $3^3 + 4^2$ _____

39 $5^2 \times 2^2$ _____

40 $3^2 \times 5^2$ _____

41 $2^3 \times 2^2$ _____

42 $3^3 + 3^2$ _____

Easy

Questions 1–22

Evaluate each square root: which number, squared, gives this?



WE DO With your teacher: copy the working.

E1 $\sqrt{49}$ _____

E2 $\sqrt{100}$ _____

YOU DO Your turn.

1 $\sqrt{169}$ _____

13

2 $\sqrt{4}$ _____

3 $\sqrt{196}$ _____

4 $\sqrt{225}$ _____

5 $\sqrt{25}$ _____

6 $\sqrt{324}$ _____

7 $\sqrt{361}$ _____

8 $\sqrt{400}$ _____

9 $\sqrt{289}$ _____

10 $\sqrt{144}$ _____

11 $\sqrt{81}$ _____

12 $\sqrt{9}$ _____

13 $\sqrt{36}$ _____

Evaluate each cube root: which number, cubed, gives this?



WE DO With your teacher: copy the working.

E3 $\sqrt[3]{8}$ _____

YOU DO Your turn.

14 $\sqrt[3]{64}$ _____

4

15 $\sqrt[3]{27}$ _____

16 $\sqrt[3]{1000}$ _____

17 $\sqrt[3]{125}$ _____

18 $\sqrt[3]{216}$ _____

19 $\sqrt[3]{729}$ _____

20 $\sqrt[3]{343}$ _____

21 $\sqrt[3]{512}$ _____

22 $\sqrt[3]{1}$ _____

Medium

Questions 23–32

Between which two whole numbers is the square root? Use the squares either side.



WE DO With your teacher: copy the working.

E1 $\sqrt{47}$ _____

E2 $\sqrt{89}$ _____

YOU DO Your turn.

23 $\sqrt{134}$ _____

$11^2 = 121, 12^2 = 144$

between 11 and 12

24 $\sqrt{42}$ _____

$6^2 = 36, 7^2 = 49$

between ____ and ____

25 $\sqrt{137}$ _____

____² = ____, ____² = ____

between ____ and ____

26 $\sqrt{28}$ _____

27 $\sqrt{51}$ _____

28 $\sqrt{32}$ _____

29 $\sqrt{92}$ _____

30 $\sqrt{11}$ _____

31 $\sqrt{135}$ _____

32 $\sqrt{141}$ _____

Challenging

Questions 33–39

Evaluate. Roots and powers first.



WE DO With your teacher: copy the working.

E1 $\sqrt{121} + \sqrt[3]{125} \times 4$

E2 $\sqrt{100} + \sqrt[3]{64} \times 2$

YOU DO Your turn.

33 $\sqrt{49} + \sqrt[3]{125} \times 5$

$= 7 + 5 \times 5$

$= 7 + 25$

$=$ ____

34 $\sqrt{9} + \sqrt[3]{8} \times 4$

$=$ ____ + ____ $\times$ ____

$=$ ____ + ____

$=$ ____

35 $\sqrt{4} + \sqrt[3]{125} \times 4$

36 $\sqrt{4} + \sqrt[3]{8} \times 2$

37 $\sqrt{144} + \sqrt[3]{64} \times 6$

38 $\sqrt{144} + \sqrt[3]{8} \times 6$

39 $\sqrt{100} + \sqrt[3]{125} \times 3$

Easy

Questions 1–29

Prime (P) or composite (C)?
A prime has exactly two factors.



WE DO With your teacher: copy the working.

E1 80

E2 19

YOU DO Your turn.

1 41

P

2 54

3 91

4 99

5 82

6 4

7 64

8 85

9 42

10 90

11 78

12 63

13 45

14 57

15 40

16 83

17 21

18 38

19 3

20 51

21 75

22 76

23 12

24 10

25 11

26 50

27 56

28 96

29 88

Medium

Questions 30–39



List all the factors in pairs.

WE DO With your teacher: copy the working.

E1 52

E2 38

YOU DO Your turn.

30 30

$1 \times 30, 2 \times 15, 3 \times 10, 5 \times 6$

factors: 1, 2, 3, 5, 6, 10, 15, 30

31 46

$1 \times 46, 2 \times 23$

factors: __, __, __, __

32 21

__ $\times$ __, __ $\times$ __

factors: __, __, __, __

33 50

34 34

35 12

36 44

37 20

38 24

39 32

Challenging

Questions 40–49

Test each number for primes:
try dividing by 2, 3, 5 and 7.



WE DO With your teacher: copy the working.

E1 Is 63 prime?

E2 Is 55 prime?

YOU DO Your turn.

40 Is 77 prime?

$77 \div 7 = 11$

composite: __ = __ $\times$ __

41 Is 75 prime?

__ $\div$ __ = __

composite: __ = __ $\times$ __

42 Is 137 prime?

43 Is 93 prime?

44 Is 83 prime?

45 Is 67 prime?

46 Is 73 prime?

47 Is 131 prime?

48 Is 105 prime?

49 Is 135 prime?

Easy

Questions 1–19

Write each number as a product of two prime numbers.



WE DO With your teacher: copy the working.

E1 49

E2 65

YOU DO Your turn.

1 143

11×13

2 39

3 77

4 14

5 169

6 121

7 55

8 35

9 10

10 15

11 21

12 91

13 25

14 22

15 4

16 6

17 26

18 9

19 33

Medium

Questions 20–26

Divide by primes until you reach 1. Write the answer in index notation.



WE DO With your teacher: copy the working.

E1 40

E2 36

YOU DO Your turn.

20 24

$$24 \div 2 = 12, 12 \div 2 = 6$$

$$6 \div 2 = 3, \dots$$

$$24 = 2^3 \times 3$$

21 42

$$42 \div 2 = 21, 21 \div 3 = 7$$

$$7 \div 7 = 1$$

$$\underline{\quad} = \underline{\quad} \times \underline{\quad} \times \underline{\quad}$$

22 70

$$\underline{\quad} \div \underline{\quad} = \underline{\quad}, \underline{\quad} \div \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} \div \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} = \underline{\quad} \times \underline{\quad} \times \underline{\quad}$$

23 30

24 20

25 60

26 18

Challenging

Questions 27–33

Use prime factors to find the root. Pair up (or group) the factors.



WE DO With your teacher: copy the working.

E1 $\sqrt{36}$

E2 $\sqrt[3]{216}$

YOU DO Your turn.

27 $\sqrt{324}$

$$324 = 2^2 \times 3^4$$

$$= (2 \times 3 \times 3)^2$$

$$\sqrt{\quad} = \underline{\quad}$$

28 $\sqrt[3]{1\,000}$

$$\underline{\quad} = \underline{\quad}^3 \times \underline{\quad}^3$$

$$= (\underline{\quad} \times \underline{\quad})^3$$

$$\sqrt[3]{\quad} = \underline{\quad}$$

29 $\sqrt{900}$

30 $\sqrt[3]{1\,728}$

31 $\sqrt{484}$

32 $\sqrt[3]{3\,375}$

33 $\sqrt{225}$

3.11 Highest common factor

YEAR 7 · CHAPTER 3: WHOLE NUMBERS

Easy

Questions 1–29

Find the highest common factor (HCF): the largest number that divides both.



WE DO With your teacher: copy the working.

E1 8 and 10

E2 24 and 20

YOU DO Your turn.

1 27 and 18

9

2 24 and 12

3 12 and 21

4 4 and 14

5 45 and 18

6 30 and 25

7 10 and 6

8 9 and 54

9 21 and 7

10 8 and 4

11 42 and 36

12 8 and 24

13 30 and 36

14 4 and 8

15 12 and 4

16 4 and 6

17 12 and 42

18 8 and 12

19 12 and 18

20 2 and 14

21 2 and 10

22 12 and 14

23 12 and 15

24 21 and 15

25 56 and 40

26 14 and 2

27 45 and 9

28 16 and 24

29 8 and 48

Medium

Questions 30–36

List the factors of each number, then circle the highest common factor.



WE DO With your teacher: copy the working.

E1 16 and 32

E2 16 and 24

YOU DO Your turn.

30 10 and 4

10: 1, 2, 5, 10

4: 1, 2, 4

HCF = 2

31 30 and 25

30: 1, 2, 3, 5, 6, 10, 15, 30

25: 1, 5, 25

HCF = 5

32 8 and 12

8: 1, 2, 4, 8

12: 1, 2, 3, 4, 6, 12

HCF = 4

33 4 and 10

34 6 and 10

35 36 and 30

36 9 and 6

Challenging

Questions 37–43

Use prime factors: the HCF is the product of the primes they share.



WE DO With your teacher: copy the working.

E1 30 and 75

E2 40 and 50

YOU DO Your turn.

37 36 and 48

36 = $2 \times 2 \times 3 \times 3$

48 = $2 \times 2 \times 2 \times 2 \times 3$

HCF = $2 \times 2 \times 3 = 12$

38 50 and 60

50 = $2 \times 5 \times 5$

60 = $2 \times 2 \times 3 \times 5$

HCF = $2 \times 5 = 10$

39 72 and 54

40 30 and 20

41 75 and 45

42 45 and 60

43 75 and 105

3.12 Lowest common multiple

Easy

Questions 1–29

Find the lowest common multiple (LCM): the smallest number both go into.

WE DO With your teacher: copy the working.

E1 10 and 5 _____

E2 11 and 8 _____

YOU DO Your turn.

1 7 and 9 63

2 3 and 9 _____

3 12 and 11 _____

4 5 and 11 _____

5 8 and 6 _____

6 8 and 4 _____

7 11 and 6 _____

8 7 and 6 _____

9 7 and 2 _____

10 10 and 6 _____

11 11 and 2 _____

12 10 and 4 _____

13 6 and 8 _____

14 4 and 5 _____

15 6 and 11 _____

16 8 and 9 _____

17 3 and 5 _____

18 8 and 2 _____

19 6 and 9 _____

20 2 and 7 _____

21 3 and 11 _____

22 12 and 2 _____

23 2 and 8 _____

24 7 and 8 _____

25 4 and 7 _____

26 3 and 2 _____

27 8 and 11 _____

28 4 and 3 _____

29 8 and 3 _____

Medium

Questions 30–36

List multiples of each number until you find one in both lists.

WE DO With your teacher: copy the working.

E1 3 and 12 _____

E2 6 and 5 _____

YOU DO Your turn.

30 3 and 6

3: 3, 6

6: 6

LCM = 6

31 6 and 12

6: 6, 12

12: 12

LCM = ____

32 12 and 6

____: ____

____: ____

LCM = ____

33 3 and 4

34 4 and 3

35 10 and 8

36 8 and 12

Challenging

Questions 37–39

Use the LCM to answer the question.

WE DO With your teacher: copy the working.

E1 One bus leaves every 12 minutes, another every 7 minutes. Both leave at 8:00. After how many minutes do they next leave together?

E2 One bus leaves every 7 minutes, another every 11 minutes. Both leave at 8:00. After how many minutes do they next leave together?

YOU DO Your turn.

37 One bus leaves every 10 minutes, another every 14 minutes. Both leave at 8:00. After how many minutes do they next leave together?

LCM of 10 and 14

= 70

after ____ minutes

38 One bus leaves every 6 minutes, another every 7 minutes. Both leave at 8:00. After how many minutes do they next leave together?

LCM of ____ and ____

= ____

after ____ minutes

39 One bus leaves every 5 minutes, another every 10 minutes. Both leave at 8:00. After how many minutes do they next leave together?